though we can successfully link to this string, there is no defined function to parse "7-5, 6-5" as "won two games" because it requires linguistic/mathematical inference to understand the implication from the string. Such cases are the weakness of symbolic reasoning models.

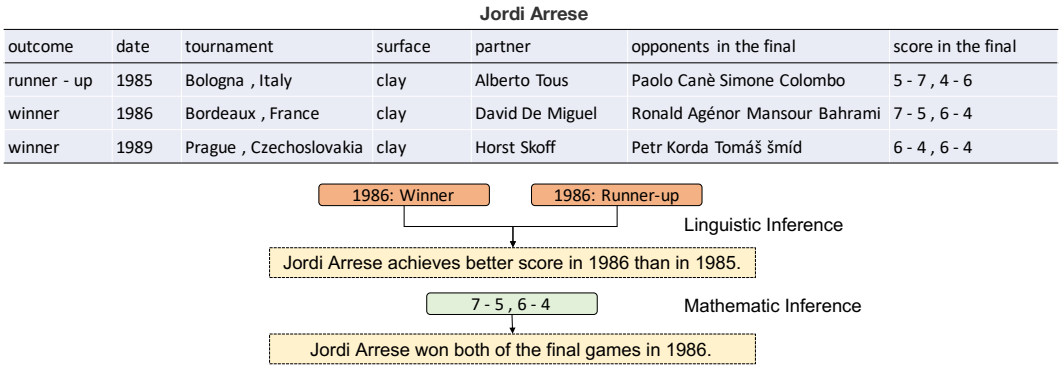

Figure 9: The error case of symbolic reasoning model

BERT In contrast, Table-BERT model seems to have no coverage problem as long as it can feed the whole table content. However, due to the template linearization, the table is unfolded into a long sequence as depicted in Figure 10. The useful information, "clay" are separated in a very long span of unrelated words. How to grasp such a long dependency and memorize the history information poses a great challenge to the Table-BERT model.

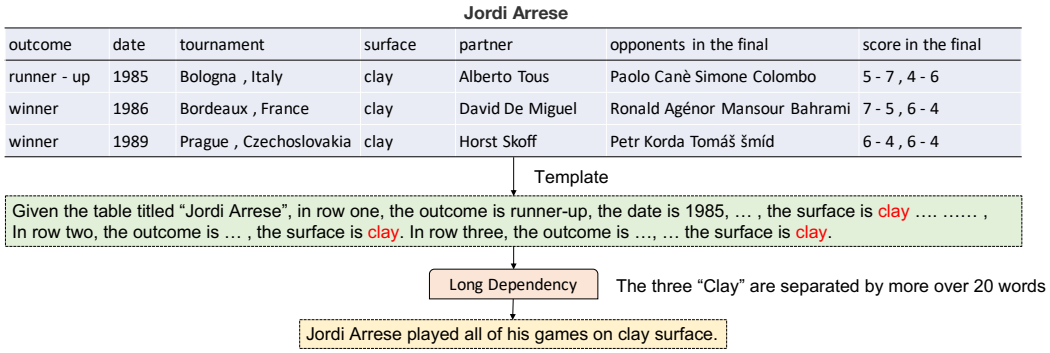

Figure 10: The error case of BERT NLI model

Statistics Here we pick 200 samples from the validation set which only involve single semantic and divide them into different categories. We denote the above-mentioned cases as "linguistic inference", and the sentences which only describe information from one row as "Trivial", the rest are based on their logic operation like Aggregation, Superlative, Count, etc. We visualize the accuracy of LPA and Table-BERT in Figure 11. From which we can observe that the statements with linguistic inference are much better handled with the BERT model, while LPA achieves an accuracy barely higher than a random guess. The BERT model can deal with trivial cases well as it uses a horizontal scan order. In contrast, the LPA model outperforms BERT on higher-order logic cases, especially when the statement involves operations like Count and Superlative.

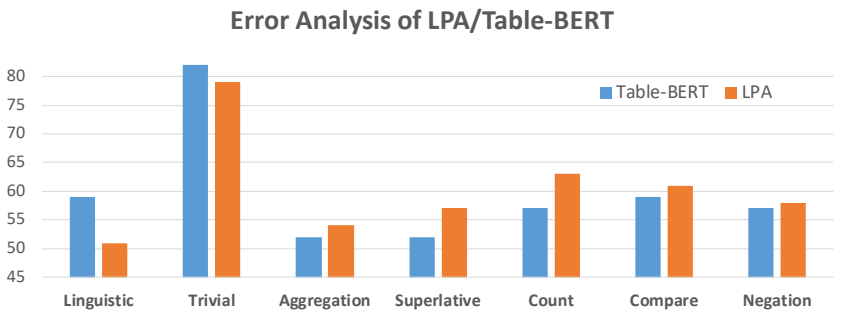

Figure 11: The error analysis of two different models

D REASONING DEPTH

Given that our LPA has the breadth to cover a large semantic space. Here we also show the reasoning depth in terms of how many logic inference steps are required to tackle verify the given claims. We visualize the histogram in Figure 12 and observe that the reasoning steps are concentrated between 4 to 7. Such statistics indicate the difficulty of fact verification in our TABFACT dataset.

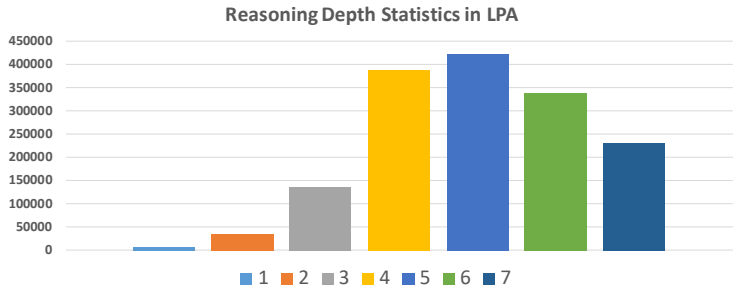

Figure 12: The histogram of reasoning steps required to verify the claims

E WHETHER TO KEEP WIKIPEDIA CONTEXT

Before crowd-sourcing the annotation for the tables, we observed that the previous WikiTableQuestion Pasupat & Liang (2015) provides context (Wikipedia title) during annotation while the WikiSQL Zhong et al. (2017) does not. Therefore, we particularly design ablation annotation tasks to compare the annotation quality between w/ and w/o Wikipedia title as context. We demonstrate a typical example in Figure 13, where a Wiki table[10] aims to describe the achievements of a tennis player named Dennis, but itself does not provide any explicit hint about "Tennis Player Dennis". Unsurprisingly, the sentence fluency and coherence significantly drop without such information. Actually, a great portion of these Wikipedia tables requires background knowledge (like sports, celebrity, music, etc) to understand. We perform a small user study to measure the fluency of annotated statements. Specifically, we collected 50 sentences from both annotation w/ and w/o title context and randomly shuffle them as pairs, which are distributed to the 8 experts without telling them their source to compare the language fluency. It turns out that the experts ubiquitously agree that the statements with Wikipedia titles are more human-readable. Therefore, we argue that such a context is necessary for annotators to understand the background knowledge to write more fluent sentences. On the other end, we also hope to minimize the influence of the textual context in the table-based verification task, therefore, we design an annotation criterion: the Wikipedia title

[10] `https://en.wikipedia.org/wiki/Dennis_Ralston`

is provided to the workers during the annotation, but they are explicitly banned from bringing any unrelated background information other than the title into the annotation. As illustrated in Figure 13, the title only acts as a placeholder in the statements to make it sound more natural.

outcome	year	championship	surface	partner
winner	1960	Wimbledon championships	grass	Rafael Osuna
winner	1961	US Championships	grass	Chuck Mckinley
runner - up	1962	US Championships	grass	Chuck Mckinley
winner	1963	US Championships (2)	grass	Chuck Mckinley

Context (Title)	*Richard Dennis Ralston (born July 27, 1942, an American former tennis player*	*No Information is provided*
Annotate	From 1960 to 1969, Ralston won five major double championships.	Winner is on the grass surface. Rafael Osuna is partner in the Wimbeldon

Figure 13: Comparison of worker annotation w/ and w/o Wikipedia title as context

F ENTITY LINKING

Here we propose to use the longest string match to find all the candidate entities in the table, when multiple candidates coexist, we select the one with the minimum edit distances. The visualization is demonstrated in Figure 14.

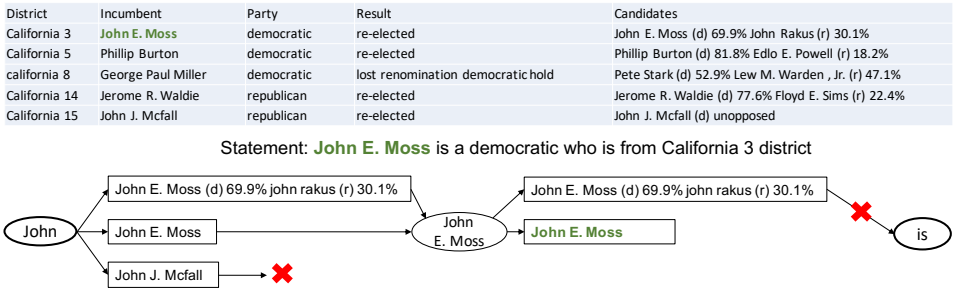

Figure 14: Entity Linking System.

G THE PROGRAM CANDIDATES

Here we demonstrate some program candidates in Figure 15, and show how our proposed discriminator is designed to compute the matching probability between the statement and program. Specifically, we employ two transformer-based encoder Vaswani et al. (2017), the left one is aimed to encode the program sequence and the right one is aimed to encode the statement sequence. Their output from [CLS] position is concatenated and fed into an MLP to classify the verification label.

H HIT INTERFACE

We provide the human intelligent task interface on AMT in the following. Very detailed instructions on what are trivial statements and what are non-trivial statements. Comprehensive examples have been given to guide the Turkers to write well-formed while logically plausible statements. In order to harvest fake statements without statistical cues, we also provide detailed instructions on how to re-write the ”fake” statements. During the annotation, we hire 8 experts to perform sanity checks on each of the HIT to make sure that the annotated dataset is clean and meets our requirements.